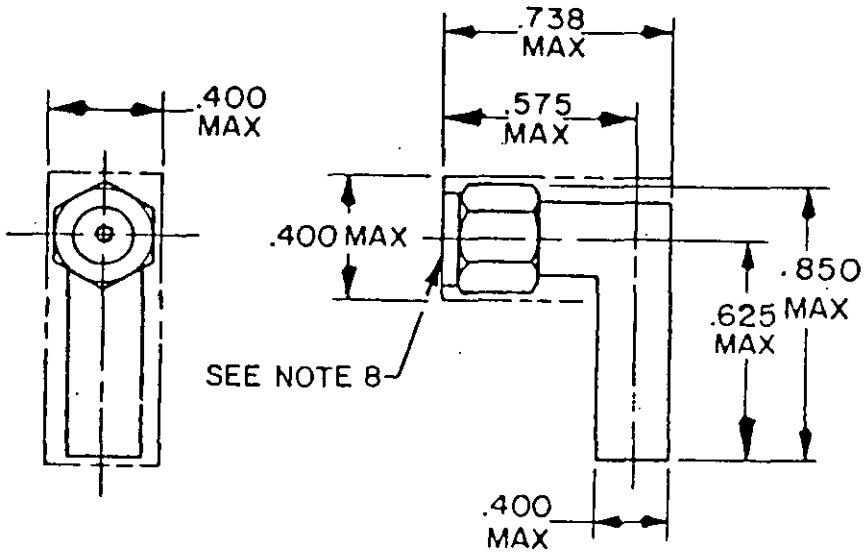
MILITARY SPECIFICATION SHEET

CONNECTORS, PLUG, ELECTRICAL, COAXIAL, RADIO FREQUENCY, SERIES SMA (CABLED, PIN RIGHT ANGLE, CLASS 2, SEMIRIGID CABLE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-39012.



Inches	mm
.016	0.41
.150	3.81
.400	10.16
.575	14.60
.625	15.88
.738	18.75
.850	21.59

MARKING IMPLEMENTATION DATE, CATEGORY B, SEE TABLE VI

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension .400 (10.16 mm) is the maximum envelope dimension.
4. Wrench flats to accommodate standard wrench opening per FED-STD-H28, appendix 10.
5. All undimensioned pictorial configurations are for reference purposes only.
6. Dimension .850 (21.59 mm) maximum defines the overall length of connector when assembled to the cable.
7. When applicable, three holes .016 (0.41 mm) minimum diameter, equally spaced, are required for safety wiring after mating. Location on the coupling nuts optional.
8. Series SMA, pin contact interface in accordance with MIL-STD-348.

FIGURE 1. General configuration.

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TABLE I. Dash number and applicable cable.

Dash no.	Applicable cable 7/
CATEGORY A - FIELD SERVICEABLE (NO SPECIAL TOOLS REQUIRED) 1/ 8/	
3009 3109 3/ 4009 4109 3/	M17/133-RG405* M17/133-00001 M17/133-00002 M17/133-00003 M17/133-00004 M17/133-00005
3010 3110 3/ 4010 4110 3/	M17/130-RG402* M17/130-00001 M17/130-00002 M17/130-00003
CATEGORY E - FIELD REPLACEABLE (STANDARD ASSEMBLY TOOL KIT) 2/ 4/ 8/ 9/	
3005 6/ 3105 3/ 6/	M17/133-RG405* M17/133-00001 M17/133-00002 M17/133-00003 M17/133-00004 M17/133-00005
3006 6/ 3106 3/ 6/	M17/130-RG402* M17/130-00001 M17/130-00002 M17/130-00003
3007 1/ 3107 1/ 3/ 4007 1/ 4107 1/ 3/	M17/133-RG405* M17/133-00001 M17/133-00002 M17/133-00003 M17/133-00004 M17/133-00005
3008 1/ 3108 1/ 3/ 4008 1/ 4108 1/ 3/	M17/130-RG402* M17/130-00001 M17/130-00002 M17/130-00003

TABLE I. Dash number and applicable cable - Continued.

Dash no.	Applicable cable 7/	Tool no.	Positioning dies	Locator pins 5/	Crimp dies M22520/5
CATEGORY F - FIELD REPLACEABLE (MIL-C-22520 CRIMP TOOL) 1/ 4/ 8/					
3207 10/ 3307 3/ 10/ 4207 10/ 4307 3/ 10/	M17/133-RG405* M17/133-00001 M17/133-00002 M17/133-00003 M17/133-00004 M17/133-00005	M22520/36-01	M22520/36-15	M22520/36-15	
3208 10/ 3308 3/ 10/ 4208 10/ 4308 3/ 10/	M17/130-RG402* M17/130-00001 M17/130-00002 M17/130-00003		M22520/36-15		---
3210 3310 3/ 4210 4310 3/	M17/130-RG402* M17/130-00001 M17/130-00002 M17/130-00003	M22520/5-01	---	---	-05 or -41 cavity B

See footnotes at end of table.

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* Cable to be used when performing tests requiring cable.

1/ These connectors have captivated center contacts.

2/ Kit number - Ommi Spectra T-200, Amphenol 901-2500, or equivalent.

3/ No safety wire holes.

4/ For logistics purposes, only connectors with safety wire holes will be stocked.

5/ The locators required shall be indicated in the assembly instructions.

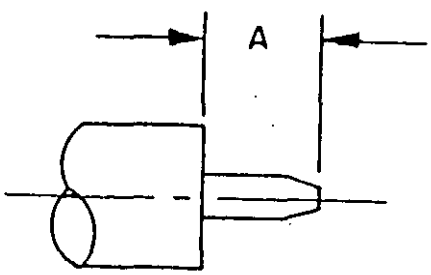
6/ Inactive for new design. Not for Air Force or Navy use.

7/ The RG cables are specified with the basic number. The latest version of each cable shall be applicable.

8/ Coupling nuts shall be corrosion-resistant steel with a passivated finish per MIL-F-14072.
(Applies to "-3XXX" series connectors only).

9/ All corrosion-resistant steel-bodied connectors shall be gold-plated per MIL-G-45204, type II, class 1.

10/ Not to be used in Army equipment.



CABLE	DIMENSION "A"	
	MIN	MAX
M17/130-RG402 M17/130-00001 M17/130-00002 M17/130-00003	.080 (2.03 mm)	.090 (2.29 mm)
M17/133-RG405 M17/133-00001 M17/133-00002 M17/133-00003 M17/133-00004 M17/133-00005	.065 (1.65 mm)	.075 (1.91 mm)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tapering of center conductor is optional.

FIGURE 2. Cable stripping dimensions for category E and category F connectors.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 12.4 GHz.

Voltage rating: The voltage rating shall be in accordance with table II.

TABLE II. Voltage rating.

Cable	Voltage max (at sea level)	Voltage max (70,000 ft)
	V rms	V rms
M17/133-RG405 M17/133-00001 M17/133-00002 M17/133-00003 M17/133-00004 M17/133-00005	335	85
M17/130-RG402 M17/130-00001 M17/130-00002 M17/130-00003	500	125

Temperature rating: -65°C to +105°C.

Category F: Connectors using semirigid cables with standard stripping dimensions and using standard military assembly tools. The method of assembly of the connector to the cable shall be solderless.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: 2 inch-pounds maximum.

Coupling proof torque: 15 inch-pounds minimum.

Inspection conditions: For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 7 to 10 inch-pounds.

Mating characteristics: See MIL-STD-348.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302, test condition B, MIL-STD-202; 5,000 megohms minimum.

Center contact retention: 6 pounds minimum axial force. Applicable to captivated-center-contact connectors only.

Radial torque: Not applicable.

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Corrosion (salt spray): Method 101, test condition B, MIL-STD-202.

Voltage standing wave ratio (VSWR): From 0.5 to 18 GHz, or approximately 80 percent of the upper cutoff frequency of the cable, whichever is lower.

Cables	VSWR
M17/133-RG405 M17/133-00001 M17/133-00002 M17/133-00003 M17/133-00004 M17/133-00005	1.10 +0.01 F (GHz)
M17/130-RG402 M17/130-00001 M17/130-00002 M17/130-00003	1.10 +0.01 F (GHz)

Swept frequency VSWR test setup:

Item 6: VSWR shall be less than 1.025 +.002 F (F in GHz).

Item 16: VSWR shall be less than 1.025 +.002 F (F in GHz).

Second step of VSWR checkout procedure: VSWR shall be less than 1.080 +.005 F (F in GHz).

Group B inspection: Use step 5, long cable method.

Qualification and group C inspection: Use step 5, long cable method.

Connector durability:

Insertion and withdrawal force:

500 cycles minimum at 12 cycles per minute maximum.

The connector shall meet mating characteristics and force to engage and disengage requirements.

Contact resistance (in milliohms maximum):

	Initial	After environment
Center contact:	4.0	6.0
Outer contact:	2.0	Not applicable
Outer cable conductor to body:	0.5 1/	Not applicable

Dielectric withstanding voltage at sea level: Method 301 of MIL-STD-202.

Cables	V rms
M17/133-RG405 M17/133-00001 M17/133-00002 M17/133-00003 M17/133-00004 M17/133-00005	1,000
M17/130-RG402 M17/130-00001 M17/130-00002 M17/130-00003	1,500

1/ Five milliohms is permissible for passivated steel bodied connectors only.

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Vibration, high frequency: Method 204, test condition D, MIL-STD-202, except the method of mounting shall be approved by the qualifying activity. No discontinuity permitted.

Shock: Method 213, test condition I, MIL-STD-202.

Thermal shock: Method 107, test condition B, MIL-STD-202, except test high temperature shall be +115°C.

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Altitude: 70,000 feet.

Cables	Volts (min)
M17/133-RG405 M17/133-00001 M17/133-00002 M17/133-00003 M17/133-00004 M17/133-00005	250
M17/130-RG402 M17/130-00001 M17/130-00002 M17/130-00003	375

RF high potential withstanding voltage:

Frequency: 5 MHz.

Leakage current: Not applicable.

Cables	V_rms
M17/133-RG405 M17/133-00001 M17/133-00002 M17/133-00003 M17/133-00004 M17/133-00005	670
M17/130-RG402 M17/130-00001 M17/130-00002 M17/130-00003	1,000

Cable retention force: The cable retention force shall be in accordance with table III.

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TABLE III. Cable retention force.

Cable	Pounds (min)	Torque 1/ inch-ounces
M17/133-RG405 M17/133-00001 M17/133-00002 M17/133-00003 M17/133-00004 M17/133-00005	30	16
M17/130-RG402 M17/130-00001 M17/130-00002 M17/130-00003	60	55

1/ Torque to be applied 4 inches maximum from the end of the connector. Reverse bend not applicable.

Coupling mechanism retention force: 60 lbs minimum.

RF leakage: -90 dB minimum tested at a frequency between 2 and 3 GHz.

This requirement may be met by meeting the RF leakage requirement on the MIL-C-39012/79 connector which has the same mating interface design.

RF insertion loss: dB max = .05 x $\sqrt{\text{freq GHz}}$. Test frequency at 9 and 12.4 GHz.

Supplemental test requirements for qualification and group C inspection for category F connectors only:

Six additional connectors shall be selected from the production lot and terminated to produce three cable leads, twelve inches minimum, sixteen inches maximum. The three assemblies shall be subjected to the following tests in sequence: VSWR, thermal shock, VSWR, and cable retention.

The following exceptions to the requirements apply to this supplemental test only:

Voltage standing wave ratio (VSWR) of the cable assembly: From 0.5 to 12.4 GHz.

Test cable assemblies	VSWR (initial)	VSWR (after thermal shock)
M17/133-RG405 M17/133-00001 M17/133-00002 M17/133-00003 M17/133-00004 M17/133-00005	1.15 +.023 F (GHz)	1.15 +.028 F (GHz)
M17/130-RG402 M17/130-00001 M17/130-00002 M17/130-00003	1.10 +.023 F (GHz)	1.10 +.025 F (GHz)

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Thermal shock: 10 cycles.

Part number: M39012/80 (dash number from table I or "B" number from table V).

TABLE IV. Group qualification.

Group	Submission and qualification of any of the following dash numbers 1/ 2/ 3/	Qualifies the following dash numbers
I	*02	*01 *02
II	*04	*03 *04
III	*06	*05 *06
IV	*08	*07 *08
V	*10	*09 *10
VI	*208	*208 *308 *207 *307
VII	*210	*210 *310

1/ Individual connectors other than listed are self qualifying only.
2/ Qualification of connectors qualifies connectors of the same material only.
3/ Connectors qualified with safety wire holes automatically qualifies connectors without safety wire holes.
* Body material and connector requirement.

NOTE: If a connector manufacturer produces a connector that meets all the requirements for two or more connector part numbers (within the same series), the manufacturer may receive qualification approval for two or more connector part numbers qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate part number. For group qualification, the connectors must be of similar design.

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TABLE V. Category B-nonfield replaceable (special tools may be required).

NOT FOR AIR FORCE OR NAVY USE. FOR OEM USE ONLY

M39012/80B 4/ 5/ 6/	Applicable cable 7/
3001 2/ 3101 3/ 2/	M17/133-RG405* M17/133-00001 M17/133-00002 M17/133-00003 M17/133-00004 M17/133-00005
3002 2/ 3102 3/ 2/	M17/130-RG402* M17/130-00001 M17/130-00002 M17/130-00003
3003 1/ 3103 1/ 3/ 4003 1/ 4103 1/ 3/	M17/133-RG405* M17/133-00001 M17/133-00002 M17/133-00003 M17/133-00004 M17/133-00005
3004 1/ 3104 1/ 3/ 4004 1/ 4104 1/ 3/	M17/130-RG402* M17/130-00001 M17/130-00002 M17/130-00003

* Cable to be used when performing tests requiring cable.

1/ These connectors have captivated center contacts.

2/ Inactive for new design.

3/ No safety wire holes.

4/ For logistics purposes, only connectors with safety wire holes will be stocked.

5/ All corrosion-resistant steel-bodied connectors which are designed to be assembled to the outer conductor using solder shall be gold-plated per MIL-G-45204, type II, class 1.

6/ Coupling nuts shall be corrosion-resistant steel with a passivated finish per MIL-F-14072.
(Applies to "-3XXX" series connectors only).

7/ The RG cables are specified with the basic number. The latest version of each cable shall be applicable.

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TABLE VI. Cross-reference of part numbers.

Replacement part number M39012/80 1/	Substitute for part number or type designation
B3003	M39012/80-3801, -3803
B4003	M39012/80-4801, -4803
B3004	M39012/80-3802, -3804
B4004	M39012/80-4804, -4802
-3007	M39012/80-3807, -3805
-4007	M39012/80-4807, -4805
-3008	M39012/80-3808, -3806
-4008	M39012/80-4808, -4806
B4103	M39012/80-4103, -4101
B4104	M39012/80-4104, -4102
B3001	M39012/80-3001
B3101	M39012/80-3101
B3002	M39012/80-3002
B3102	M39012/80-3102

1/ The new "B" part numbers will be required marking 6 months after the date of this specification. The previous part number may be used in the interim.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AM
Navy - AS, OS, MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3518-27)

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